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LETTERS: RESEARCH LETTER

**First-in-Human
Experience With a
Biventricular, Pulsatile
MCS Platform Technology
Avoiding Blood Contact**

Translational Insights



Over the last decades, the development of mechanical circulatory support (MCS) devices has nearly exclusively focused on technologies that necessitate sustained contact with blood. This holds true for technologies of both durable and temporary applications.

To overcome the complications associated with blood contact, we have developed an extravascular platform technology for univentricular or biventricular pulsatile support following extensive regulatory-guided development and preclinical testing.^{1,2} The concept is also the focus of several research groups that have reported promising results.^{3,4}

We present interim data from the first-in-human clinical feasibility trial, which was approved by the competent authorities in Germany (Bundesinstitut für Arzneimittel und Medizinprodukte), United Kingdom (Medicines and Healthcare products Regulatory Agency), and Denmark (Danish Medicines Agency) as well as the Ethics Committees of Hannover Medical School (Germany), Freeman Hospital in Newcastle upon Tyne (United Kingdom), and the Rigshospitalet in Copenhagen (Denmark), respectively. Heart failure (HF) patients gave written informed consent and received short-term application of the study device immediately before elective durable LVAD implantation via sternotomy.⁵

The implant comprises a cardiac sleeve that holds 3 inflatable cushions and 6 epicardial electrodes, as well as an electro-pneumatic driveline connected to

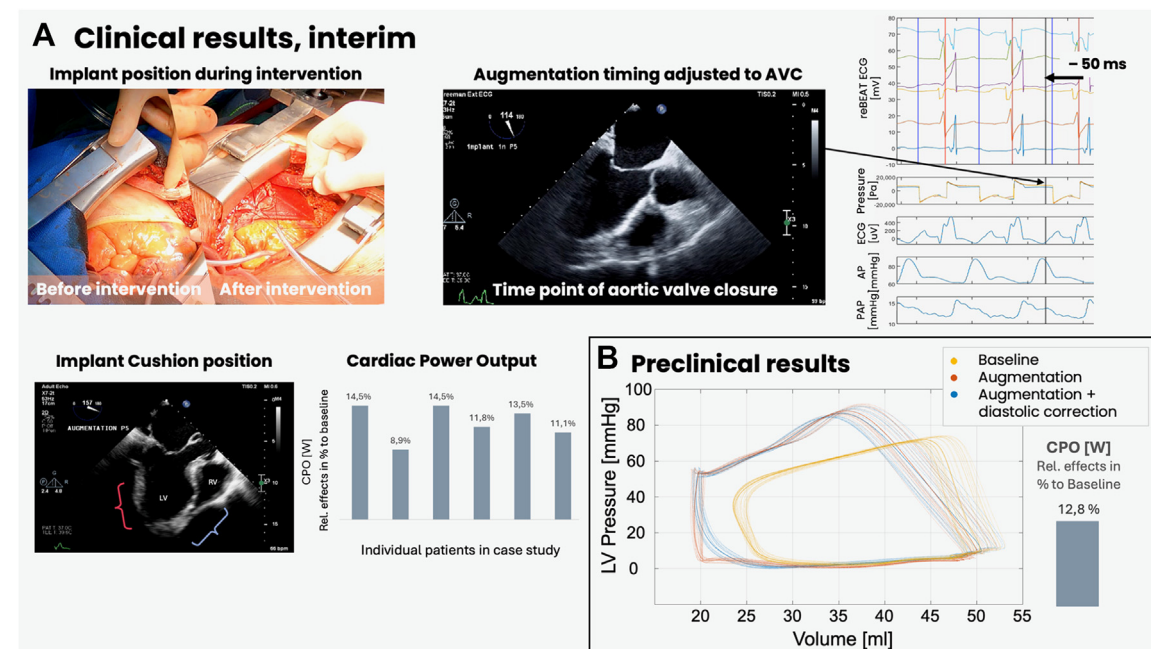
an external controller. The controller receives and interprets the epicardial electrocardiography (ECG) signals in real-time. It provides pneumatic energy to inflate and deflate the cushions in precise synchrony with the native heart cycle in a rate-adaptive manner. The pneumatic energy delivered to each cushion is individually adjustable.

A clinical workflow was established to select a patient-specific fitting implant from a set of 30 different standardized sizes (off-the-shelf) covering the majority of cardiac geometries of the study population. These standardized sizes are based on a continuously growing, anonymized cardiac computed tomography database of HF patients to sufficiently represent cardiac geometric heterogeneity. The patient-specific fit is determined by comparing a computed tomography-based 3-dimensional reconstruction of the ventricular aspect of the patient's heart at end-diastole to an in-house fitting algorithm assessing the standardized implant sizes. Patients are included in the study in agreement with the trial management group of the study. The duration from informed consent to implantation was realized within 4 to 6 days.

In 6 patients, implantation and positioning of the implant around the heart was performed in a few minutes on the beating heart and immediate detection of 6 ECG channels from around the heart was confirmed, allowing for initiation of synchronized pneumatic augmentation by the device. For each baseline and augmentation phase, 5-minute intervals were collected for data analysis.

Safety endpoints have been reached, including no-device related adverse cardiovascular events during 30-day follow-up and no new ECG abnormalities, and no harmful device-tissue interactions were detected in all patients.

Performance endpoints have been reached, including reliable detection of epicardial ECG signals to allow for synchronized augmentation. In 3 patients, the time point of pneumatic deflation was adjusted to improve diastolic filling and avoid asynchrony. No adverse effects on valve function were observed. Overall, improvement of one or more

FIGURE 1 Clinical and Preclinical Data

(A) Clinical results: implant position is shown before and after intervention. The precision of deflation was adjusted for this patient by -50 ms to optimize electromechanic synchronization verified by TOE. Cushion position is identified during TOE. Right heart cardiac power output (CPO) in response to augmentation is shown for 6 individual patients. (B) Preclinical performance of the device shows upward and leftward shift of the pressure volume loops during augmentation vs baseline. In addition, correction of the T-wave diastolic deflation point by -50 ms for this animal improves isovolumetric relaxation. Augmentation results in an increase in CPO. AP = arterial pressure; ECG = electrocardiography; LV = left ventricular; PAP = pulmonary artery pressure.

hemodynamic parameters were accomplished, including an increase in right heart cardiac power output (Figure 1).

This world's (ongoing) first-in-human study proves feasibility of this extravascular technology in advanced HF patients. Aside from an excellent safety profile, translational aspects were considered only discernable in the clinical setting when compared with the animal models studied.

We implemented the possibility to adjust the timepoint of active pneumatic deflation relative to aortic valve closure for up to ± 100 ms, as guided by transesophageal echocardiography, and adjusted it in 3 patients to optimize electromechanic synchronization to benefit overall hemodynamic efficiency.

In addition, the amount of pneumatic energy delivered into the cushions was limited intentionally for safety reasons, but higher energy levels may be applied in future patients. This demands precise adjustment of energy supply to demand and allow for

studying potential effects on atrioventricular valve function.

Moreover, the overall higher heterogeneity of the human cardiac geometries compared with the animal models revealed the fitting process to be of utmost importance, which iteratively improved over the course of this study.

This novel extravascular MCS device platform allows for precise, patient-specific adjustment of uni-ventricular or biventricular support. Customized implants may be delivered within a few days. The implantation procedure is safe and short in duration on the beating heart, allowing for fast recovery of the patient. Echocardiography can be used to evaluate patients and enhance device performance based on individual needs. Following the completion of the study, feasibility and pivotal trials are expected to introduce this new technology for temporary support in HF patients awaiting heart transplantation in the United States.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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